

IBM University Engagement Project Submission

"Exploring the Power of Agentic AI with IBM BOB and IBM Granite Models"

Project Tile – [AI Health Symptom Assessment Agent]

Domain of Project – [Healthcare]

Faculty Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement -

AI Health Symptom Assessment Agent

The Challenge

Many individuals struggle to understand the significance of their symptoms and whether they require medical attention.

The Objective

Develop an AI-powered Health Symptom Assessment Agent that interacts with users, analyzes reported symptoms, identifies possible health conditions, assesses risk levels, and provides appropriate healthcare recommendations. The solution should:

- **Analyze Symptoms** – Collect and analyze user-reported symptoms to understand their health condition.
- **Identify Possible Conditions** Predict possible health conditions based on the user's symptoms and clinical information.
- **Assess Risk Levels** – Evaluate symptom severity and classify the patient's condition as Low, Moderate, or High Risk.
- **Provide Recommendations** – Generate personalized healthcare guidance and recommend appropriate next steps based on the assessment.

Proposed Solution -

Proposed Solution: AI Health Symptom Assessment Agent

The proposed solution is an **AI-powered Health Symptom Assessment Agent** built using IBM BOB, IBM watsonx.ai, and the IBM Granite 3 H Small model to provide intelligent, real-time health assessments. The system interacts with users, analyzes their symptoms, evaluates risk levels, and provides personalized healthcare recommendations.

The solution uses Large Language Model (LLM) capabilities to understand natural language symptom descriptions, interpret medical history, and generate clinically relevant responses. The AI-powered assessment helps users better understand the seriousness of their symptoms while encouraging timely medical consultation.

The system consists of four key agents:

- **Symptom Analysis Agent** collects patient information such as age, gender, symptoms, duration, medical history, and current medications, then analyzes the reported symptoms.
- **Condition Identification Agent** compares the analyzed symptoms with medical knowledge to identify possible health conditions and presents the most relevant possibilities without making a definitive diagnosis.
- **Risk Assessment Agent** evaluates symptom severity, patient history, and identified conditions to classify the health risk as **Low, Moderate, or High**, while detecting emergency warning signs.
- **Recommendation Agent** generates personalized healthcare recommendations, including home-care advice, medical consultation, lifestyle suggestions, and emergency guidance based on the assessed risk level.

The solution provides a user-friendly web dashboard that displays patient information, clinical summaries, possible conditions, risk scores, red-flag symptoms, and healthcare recommendations in an easy-to-understand format.

IBM BOB is used to orchestrate the AI workflow, while the IBM Granite 3 H Small model performs intelligent symptom analysis and recommendation generation through natural language understanding.

Overall, the proposed solution delivers an intelligent healthcare assistant that promotes early symptom assessment, informed healthcare decisions, and timely medical intervention, helping users access reliable preliminary health guidance anytime and anywhere.

Technology Used

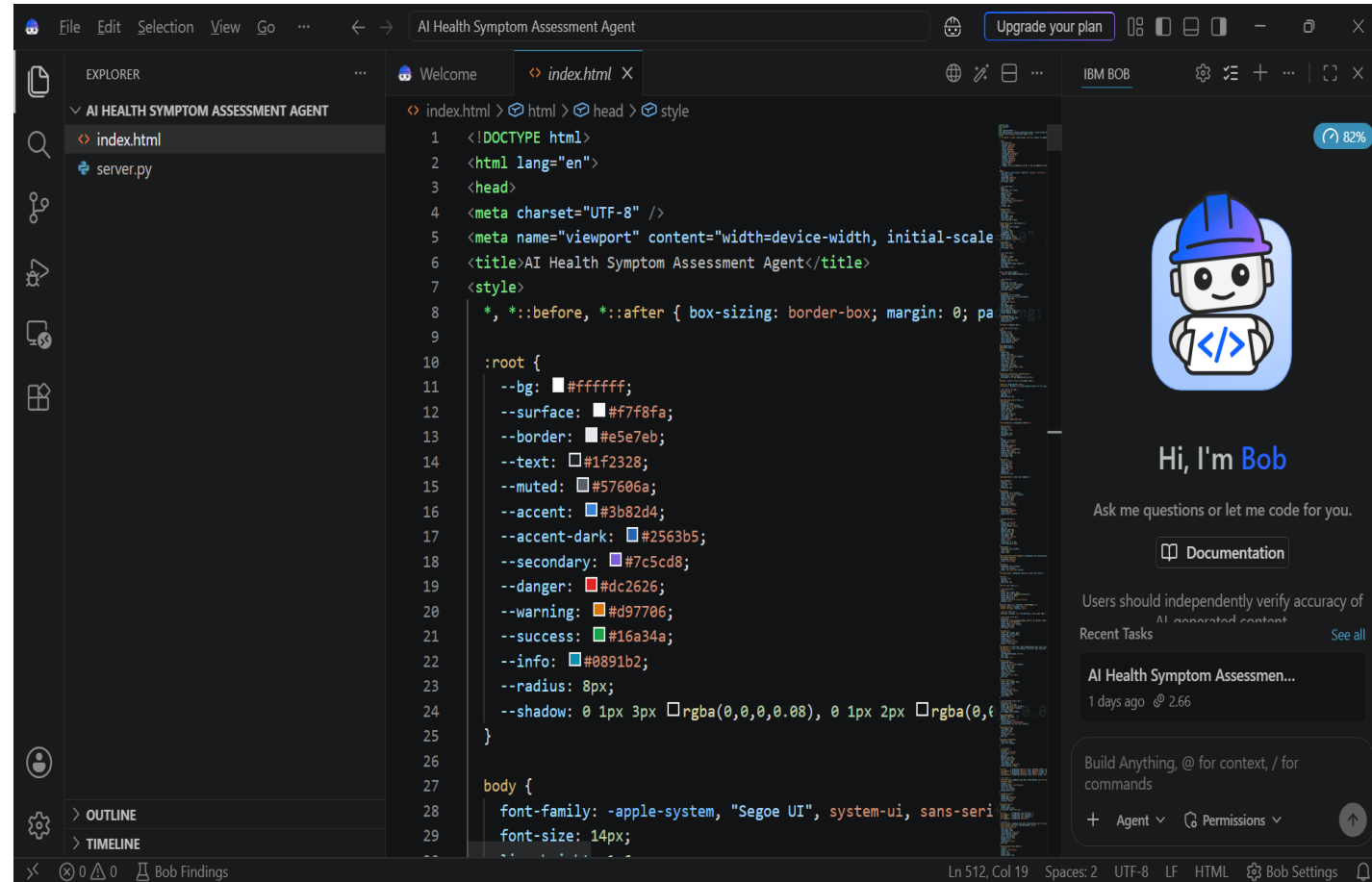
- **IBM BOB Platform** – Used to design, develop, and orchestrate the AI Health Symptom Assessment Agent workflow, enabling intelligent interaction between different AI components.
- **IBM Granite 3 H Small Model** – Provides natural language understanding, symptom analysis, clinical reasoning, and generation of personalized healthcare recommendations based on user inputs.
- **IBM Cloud** – Offers a secure, scalable, and reliable cloud infrastructure for deploying, hosting, and managing the AI Health Symptom Assessment Agent.
- **IBM watsonx.ai** – Used for accessing and deploying the IBM Granite model, enabling prompt execution, model inference, and AI-powered healthcare assessments.
- **Prompt Engineering** – Carefully designed prompts guide the Granite model to accurately analyze symptoms, assess risk levels, identify possible conditions, and generate appropriate recommendations.
- **HTML, CSS & JavaScript** – Used to build a responsive and user-friendly web interface for collecting patient information and displaying assessment reports.
- **REST API Integration** – Connects the frontend application with IBM watsonx.ai services, enabling seamless communication between the user interface and the AI model.

IBM BOB component Used -

1. **Chat Input** – Provides an interactive interface where users enter patient details such as age, gender, symptoms, symptom duration, medical history, and current medications for health assessment.
2. **IBM watsonx.ai Agent** – Processes the user input using the IBM Granite 3 H Small model to analyze symptoms, identify possible health conditions, assess risk levels, and generate healthcare recommendations.
3. **IBM Granite 3 H Small Model** – Performs natural language understanding, clinical reasoning, symptom interpretation, and generates accurate, context-aware healthcare responses in real time.
4. **Chat Output** – Displays the AI-generated assessment report, including the clinical summary, possible medical conditions, risk score, red-flag symptoms, and personalized healthcare recommendations.
5. **File Component** – Stores and retrieves patient information, symptom records, assessment reports, and prompt templates required for generating consistent and reliable AI responses.
6. **Prompt Template Component** – Uses carefully designed prompts to guide the Granite model in collecting symptom information, performing clinical analysis, classifying risk levels, and generating structured healthcare recommendations.

Required Files created using IBM Bob -

1. index.html
2. server.py



FileEditSelectionViewGo...<=>AI Health Symptom Assessment Agent

EXPLORERAI HEALTH SYMPTOM ASSESS...apiJS iam.jsJS watsonx.js.dockignoreDockerfileindex.htmlserver.pyvercel.json

api > JS iam.js > ...

```
1 // Vercel serverless function - proxies IBM IAM token requests
2 // POST /api/iam -> https://iam.cloud.ibm.com/identity/token
3 export default async function handler(req, res) {
4   res.setHeader('Access-Control-Allow-Origin', '*');
5   res.setHeader('Access-Control-Allow-Methods', 'POST, OPTIONS');
6   res.setHeader('Access-Control-Allow-Headers', 'Content-Type');
7
8   if (req.method === 'OPTIONS') {
9     return res.status(204).end();
10  }
11  if (req.method !== 'POST') {
12    return res.status(405).json({ error: 'Method not allowed' });
13  }
14
15  try {
16    const body = typeof req.body === 'string' ? req.body : JSON.stringify(req.body);
17
18    const upstream = await fetch('https://iam.cloud.ibm.com/identity/token', {
19      method: 'POST',
20      headers: { 'Content-Type': 'application/x-www-form-urlencoded' },
21      body,
22    });
23
24    const text = await upstream.text();
25    res.status(upstream.status)
26      .setHeader('Content-Type', 'application/json')
27      .send(text);
28  } catch (err) {
29    res.status(502).json({ error: err.message });
30  }
31 }
```

IBM BOB

today i worki... 38.3k / 270.0k 2.36

All tasks completed! 5/5

Ran python --version 2>&1

Python is available. Let me start the server.

Ran Start-Process -FilePath "...

Tools awaiting approval:

Execute Command

Start-Sleep -Seconds 2; try { \$r = Inv

Approve onceReject

Approve for task

Add commands to global allow/deny list

7 files changed Undo all Show all

Follow up, ↵ to steer, Alt ↵ to queue

+ Agent Permissions

master 0 0 Bob FindingsLn 1, Col 1 Spaces: 2 UTF-8 LF {} JavaScript Bob Settings

File Edit Selection View Go ...

AI Health Symptom Assessment Agent

Upgrade your plan

EXPLORER

AI HEALTH SYMPTOM ASSESS...

api

JS iam.js

JS watsonx.js

.dockerignore

Dockerfile

index.html

server.py

vercel.json

OUTLINE

TIMELINE

Welcome

JS watsonx.js

api > JS watsonx.js > ...

```
1 // Vercel serverless function - proxies IBM WatsonX chat requests
2 // POST /api/watsonx → https://us-south.ml.cloud.ibm.com/ml/v1/
3 export default async function handler(req, res) {
4   res.setHeader('Access-Control-Allow-Origin', '*');
5   res.setHeader('Access-Control-Allow-Methods', 'POST, OPTIONS');
6   res.setHeader('Access-Control-Allow-Headers', 'Content-Type, Authorization');
7
8   if (req.method === 'OPTIONS') {
9     return res.status(204).end();
10  }
11  if (req.method !== 'POST') {
12    return res.status(405).json({ error: 'Method not allowed' });
13  }
14
15  try {
16    const auth = req.headers['authorization'] || '';
17    const body = typeof req.body === 'string' ? req.body : JSON.stringify(req.body);
18
19    const upstream = await fetch(
20      'https://us-south.ml.cloud.ibm.com/ml/v1/text/chat?version=2024-08-16',
21      {
22        method: 'POST',
23        headers: {
24          'Content-Type': 'application/json',
25          'Accept': 'application/json',
26          'Authorization': auth,
27        },
28        body,
29      }
30    );
31  }
32 }
```

IBM BOB

today i worki... 38.3k / 270.0k 2.36

All tasks completed! 5/5

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7 files changed Undo all Show all

Follow up, ↵ to steer, Alt ↵ to queue

Agent Permissions

master

0 0

Bob Findings

Ln 1, Col 1 Spaces: 2 UTF-8 LF {} JavaScript Bob Settings

IBM Bob Screenshots:

File Edit Selection View Go ...

AI Health Symptom Assessment Agent

Upgrade your plan

EXPLORER

AI HEALTH SYMPTOM ASSESSMENT AGENT

index.html

server.py

OUTLINE

TIMELINE

Welcome

index.html

index.html > html > head > style

```
2 <html lang="en">
581 <body>
808 <script>
1163 /* — UI HELPERS — */
1164 function setLoading(on) {
1165     const btn = document.getElementById('assessBtn');
1166     btn.disabled = on;
1167     btn.classList.toggle('loading', on);
1168     document.getElementById('spinner').style.display = on ? 'block'
1169 }
1170 function showError(msg) {
1171     document.getElementById('errorMsg').textContent = msg;
1172     document.getElementById('errorBox').style.display = 'flex';
1173 }
1174 function hideError() { document.getElementById('errorBox').style
1175 function loadingSkeleton() {
1176     return `<div class="card"><div class="skeleton-wrap">
1177         <div class="skeleton sk-line" style="width:50%"></div>
1178         <div class="skeleton sk-block"></div>
1179         <div class="skeleton sk-line"></div>
1180         <div class="skeleton sk-line-sm"></div>
1181         <div class="skeleton sk-block"></div>
1182         <div class="skeleton sk-line"></div>
1183         <div class="skeleton sk-line-sm"></div>
1184     </div></div>`;
1185 }
1186 </script>
1187 </body>
1188 </html>
```

IBM BOB

82%



Hi, I'm Bob

Ask me questions or let me code for you.

Documentation

Users should independently verify accuracy of AI generated content

Recent Tasks

AI Health Symptom Assessmen...

1 days ago @ 2.66

Build Anything, @ for context, / for commands

+ Agent Permissions

Ln 512, Col 19

Spaces: 2

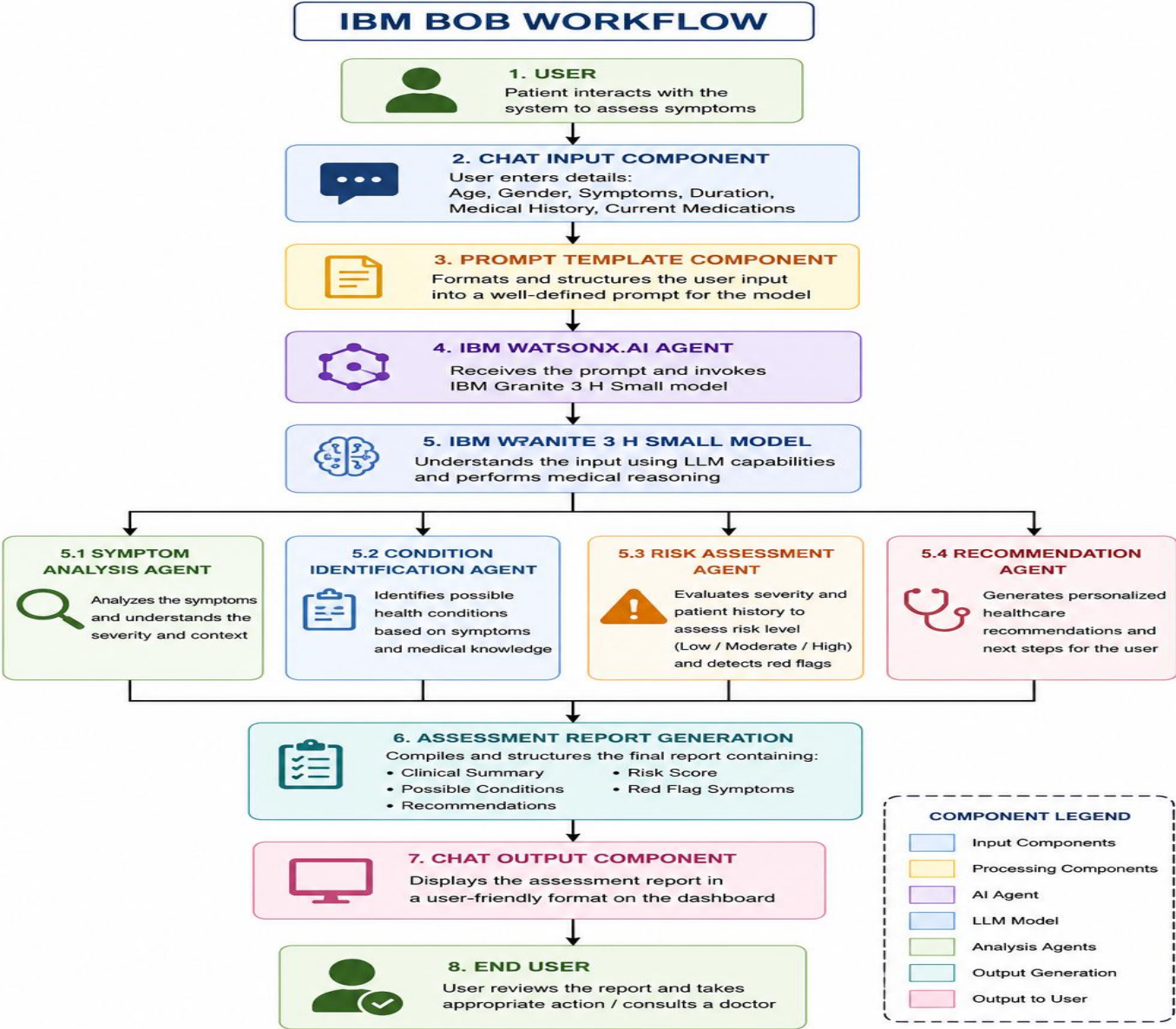
UTF-8

LF

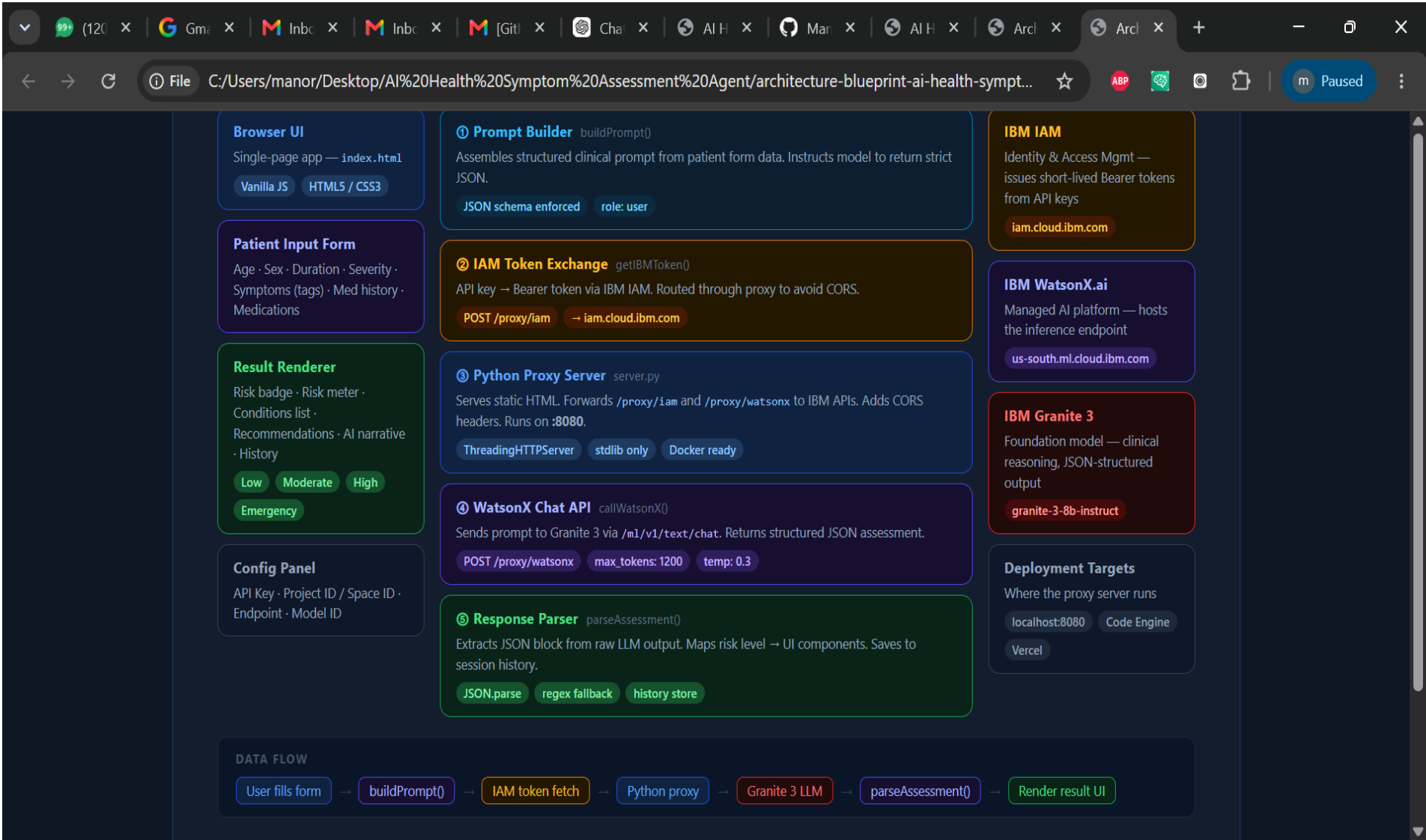
HTML

Bob Settings

IBM BOB workflow -



Architecture blueprint



Role of Agentic AI in AI Health Symptom Assessment Agent



01 **Agentic AI** enables the AI Health Symptom Assessment Agent to function as an intelligent, autonomous healthcare assistant rather than a simple symptom checker. Using **IBM watsonx.ai** and the **IBM Granite 3 H Small** model, the system understands natural language inputs, analyzes patient symptoms, and coordinates multiple AI-driven tasks to provide meaningful healthcare guidance.



02 The solution employs **specialized AI agents** that collaboratively perform symptom analysis, condition identification, risk assessment, and recommendation generation. Each agent focuses on a specific healthcare task while sharing information with the others, ensuring accurate, context-aware, and reliable assessments.



03 Agentic AI maintains **context awareness** by considering patient details such as age, gender, symptom duration, medical history, and current medications before generating responses. This enables the system to deliver personalized health assessments instead of generic advice.



04 The AI continuously performs **reasoning and decision-making** to evaluate symptom severity, identify possible health conditions, classify the patient's risk level (**Low**, **Moderate**, or **High**), and detect red-flag symptoms requiring immediate medical attention.

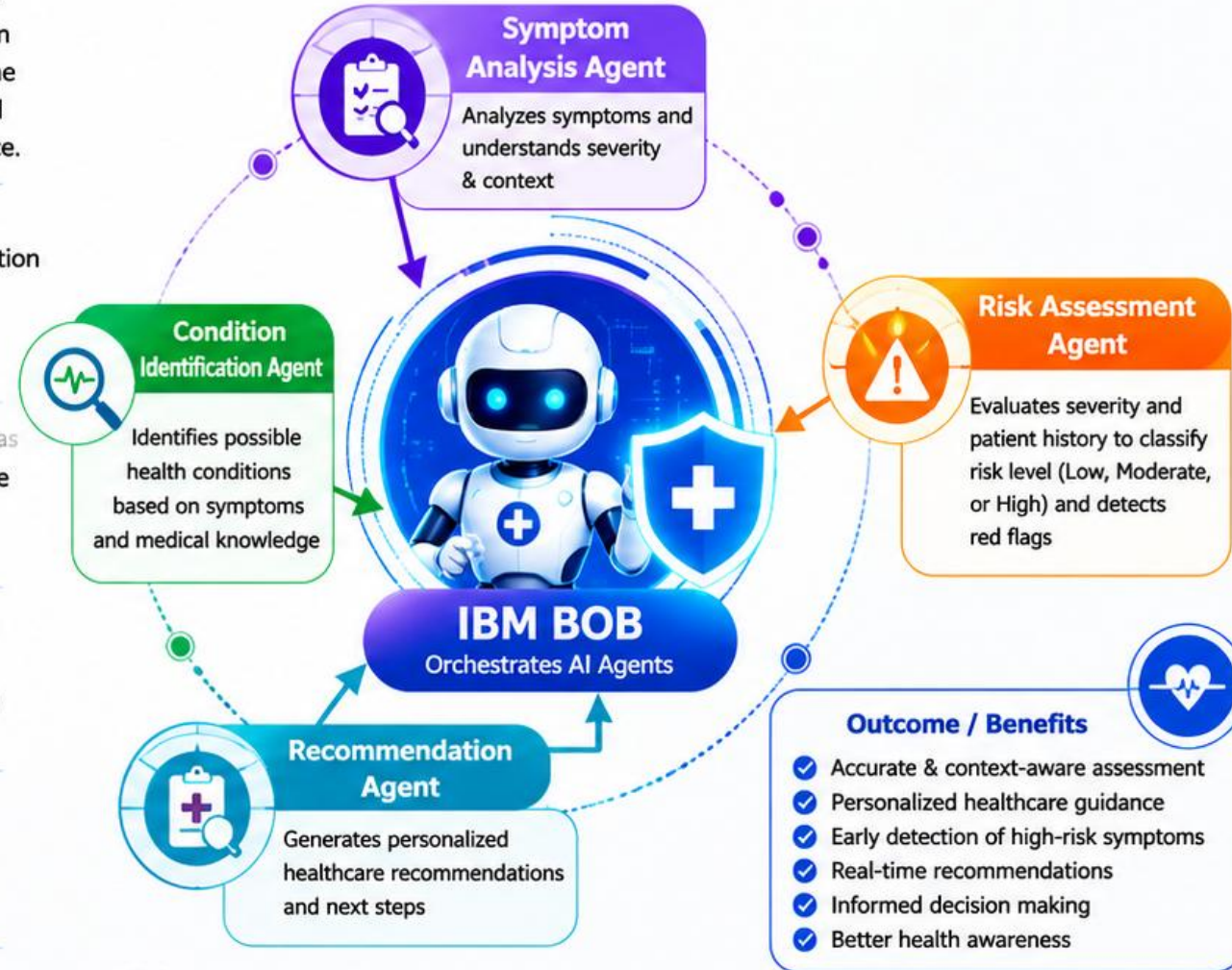


05 **IBM BOB** orchestrates the interaction between the AI agents, while **IBM Granite 3 H Small** provides advanced language understanding and medical reasoning to generate structured clinical summaries and healthcare recommendations in real time.



06 Overall, Agentic AI transforms the application into an **intelligent digital health assistant** that supports early symptom assessment, improves healthcare awareness, assists users in making informed decisions, and encourages timely consultation with qualified medical professionals.

Multi-Agent Collaboration



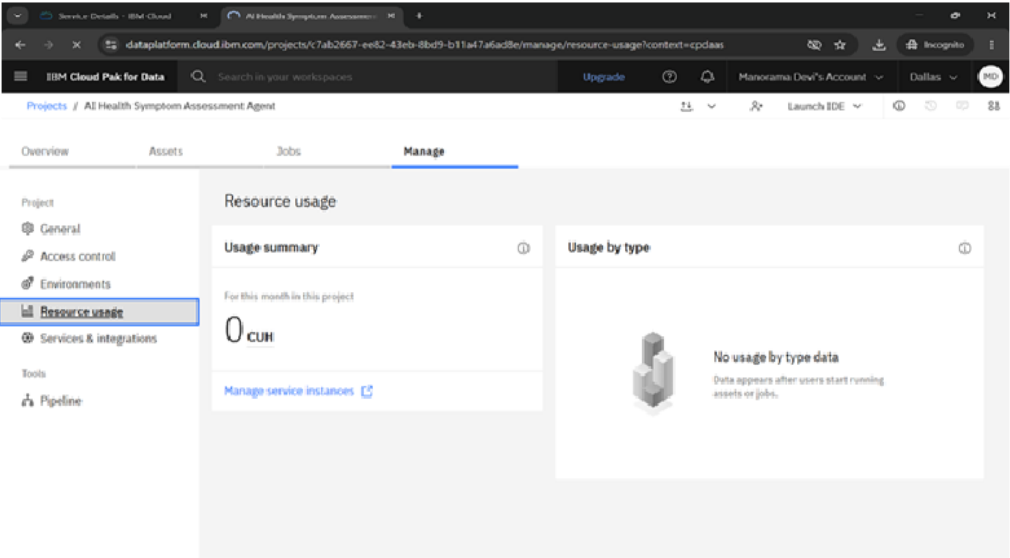
Agentic AI transforms the system into an
Intelligent Digital Health Assistant



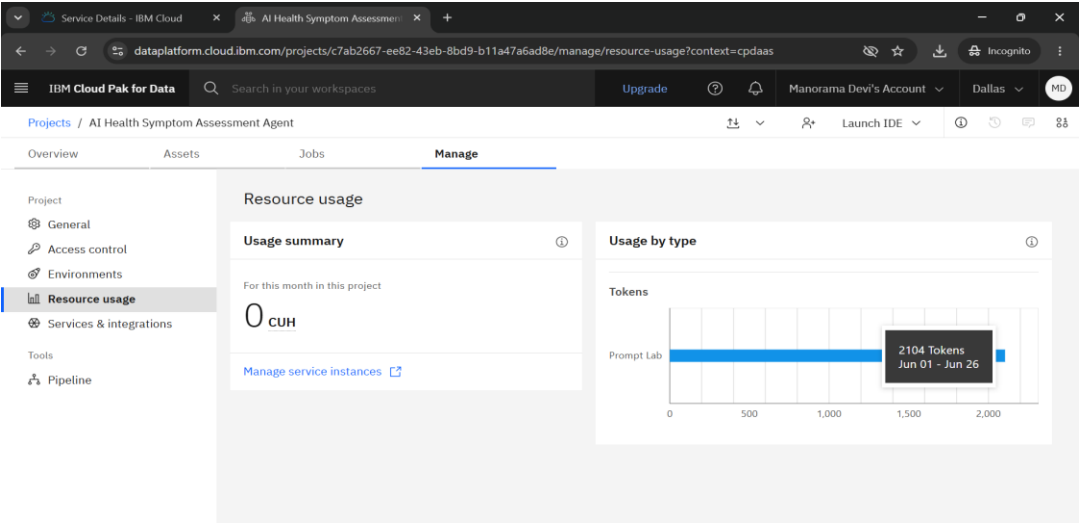
Token Usage

2104 token are used

Before Prompt



After the Prompt



Project Output Screenshot

99+ (124) W x | G Gmail: S x | M Inbox (1 x | M Inbox - x | M Inbox (7 x | Problem x | IBM Saa x | AI Healt x | AI Healt x + - x

localhost:8080

ABP

Paused

AI Health Symptom Assessment Agent IBM WATSONX Powered by IBM Granite 3 · Project c7ab2667

PATIENT INFORMATION

AGE

e.g. 35

SEX

Select

SYMPTOM DURATION

Select duration

OVERALL SEVERITY

Select severity

SYMPTOMS (add each symptom)

Type a symptom... +

Headache

Fever

Cough

Fatigue

Nausea

Chest pain

Shortness of breath

Health Symptom Assessment Agent

Powered by IBM Granite 3 on WatsonX. Enter your symptoms and patient information on the left to receive an AI-powered health assessment with risk analysis and care recommendations.

AI

AI-driven symptom analysis using IBM Granite 3 LLM

Rx

Personalised care recommendations and next steps

R

Risk stratification: Low, Moderate, High, Emergency

Dx

Possible condition identification with probability scoring

PATIENT INFORMATION

AGE

45

SEX

Female

SYMPTOM DURATION

1–3 days

OVERALL SEVERITY

Moderate – somewhat limiting daily activities

SYMPTOMS (add each symptom)

Type a symptom... +

headache x fever x cough x fatigue x

Headache Fever Cough Fatigue Nausea Chest pain Shortness of breath

Health Symptom Assessment Agent

Powered by IBM Granite 3 on WatsonX. Enter your symptoms and patient information on the left to receive an AI-powered health assessment with risk analysis and care recommendations.

AI

AI-driven symptom analysis using IBM Granite 3 LLM

R

Risk stratification: Low, Moderate, High, Emergency

Rx

Personalised care recommendations and next steps

Dx

Possible condition identification with probability scoring

99+ (125) W x | Gmail: S x | Inbox (1 x | Inbox - x | Inbox (7 x | Problem x | IBM Sa x | AI Healt x | AI Healt x | + | - | □ | x

localhost:8080 | 🔍 | ☆ | ABP | 🌐 | 📷 | 📄 | m Paused | ⋮

AI Health Symptom Assessment Agent IBM WATSONX | Powered by IBM Granite 3 · Project c7ab2667

PATIENT INFORMATION

AGE

40

SEX

Male

SYMPTOM DURATION

4-7 days

OVERALL SEVERITY

Severe – significantly limiting daily activities

SYMPTOMS (add each symptom)

Type a symptom... +

fever x

Headache

Fever

Cough

Fatigue

Nausea

Chest pain

Shortness of breath

Dizziness

Back pain

Sore throat

Abdominal pain

Joint pain

Rash

Vomiting

RELEVANT MEDICAL HISTORY

Recent surgery

CURRENT MEDICATIONS

Assessment Report

MODERATE RISK Jun 25, 04:23 PM

Risk Score

65/100

Moderate: Considering the patient's recent surgery and severe fever, there is a potential for complications or infection.

CLINICAL SUMMARY

The patient is a 40-year-old male presenting with a severe fever 4-7 days post-surgery. The risk of post-surgical infection is high, and other possible conditions include urinary tract infection and diabetic peripheral neuropathy exacerbation. Immediate medical attention is crucial to evaluate and treat any potential complications.

POSSIBLE CONDITIONS

Post-surgical infection

Infection at the surgical site, potentially caused by bacteria entering during surgery or after.

High

Diabetic peripheral neuropathy

A chronic condition caused by diabetes, leading to nerve damage that could be exacerbated by fever.

Low

Urinary tract infection

Infection in the urinary tract, which can cause fever and other symptoms.

Moderate

RED FLAG SYMPTOMS NOTED

PROLONGED FEVER (>3 DAYS)

WORSENING PAIN OR SWELLING AT SURGICAL SITE

CHILLS OR RIGORS

RECOMMENDATIONS

Novelty and Uniqueness

- The proposed solution leverages **Agentic AI** to provide an intelligent, context-aware health assessment rather than a traditional symptom checker.
- It integrates **IBM BOB**, **IBM watsonx.ai**, and the **IBM Granite 3 H Small** model for autonomous symptom analysis and healthcare guidance.
- The system considers **patient age, gender, symptom duration, medical history, and current medications** to generate personalized assessments.
- It performs **multi-step AI reasoning**, including symptom analysis, possible condition identification, risk assessment, and recommendation generation.
- The application classifies health conditions into **Low, Moderate, and High Risk**, helping users understand the urgency of their symptoms.
- It provides **real-time, personalized healthcare recommendations** instead of generic advice.
- The interactive dashboard presents **clinical summaries, possible conditions, red-flag symptoms, and recommendations** in a user-friendly format.
- The modular architecture enables future integration with **electronic health records (EHRs), telemedicine platforms, wearable devices, and other healthcare systems**, making the solution scalable and extensible.

Git Hub Link

<https://github.com/Manorama-Devi/AI-Health-Symptom-Assessment-Agent>

Files in Git Hub Repository

1. AI Health Symptom Assessment Agent Project Code.pdf
2. AI Health Symptom Assessment Agent - Problem Statement.pdf
3. IBM BOB Project PPT B. Manorama Updated.pptx

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Future Scope

- Voice input
- RAG integration
- Appointment booking
- Multi-agent collaboration

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Certificates Earned - Please add your credly certificate



Certificates Earned - Add your IBM BOB Certificate

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to
manorama devi

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 23 Jun 2026 (GMT)

Learning hours: 30 mins

IBM Skills Build Profile

The screenshot shows a web browser window with multiple tabs. The active tab is 'Profile | SkillsBuild' with the URL 'skills.yourlearning.ibm.com/profile/preference'. The page header includes the 'IBM SkillsBuild' logo and navigation links for 'Recommendations', 'Progress', a search icon, a help icon, and a user profile icon labeled 'MD'. The main content area is titled 'Profile' and features a circular profile picture with the initials 'MD'. To the right of the picture, the name 'manorama devi' is displayed with a blue edit icon, followed by the email 'manoramachithaluru@gmail.com', the location 'Andhra Pradesh, India', and the affiliation 'ACA - Edunet Foundation'. Below the profile information, the section 'Areas of interest' is introduced with the text 'Identify your areas of interest to see the most relevant promotions and recommendations.' This section is divided into two categories: 'Job roles' and 'Professional skills'. Under 'Job roles', there are eight selectable tags: 'Customer service representative', 'Cybersecurity analyst', 'Data analyst', 'Designer', 'Enterprise computing professional', 'IT support technician', 'Project manager', and 'UX designer'. Under 'Professional skills', there are six selectable tags: 'Career planning', 'Collaboration', 'Communication', 'Critical thinking', 'Decision making', and 'Digital literacy'. The bottom of the page shows the beginning of another 'Professional skills' section with additional tags partially visible.

Profile | SkillsBuild

Generative AI in Action (| Credly by Pearson | Share Badge - Credly | IBMDesign20260623-32

skills.yourlearning.ibm.com/profile/preference

Incognito (2)

IBM SkillsBuild

Recommendations Progress

Profile

 manorama devi 
manoramachithaluru@gmail.com
Andhra Pradesh, India
ACA - Edunet Foundation

Areas of interest

Identify your areas of interest to see the most relevant promotions and recommendations.

Job roles

Customer service representative Cybersecurity analyst Data analyst Designer Enterprise computing professional IT support technician Project manager
UX designer Web developer

Professional skills

Career planning Collaboration Communication Critical thinking Decision making Digital literacy Diversity and inclusion Interpersonal skills

Thank You!

Thank you for your time and interest.